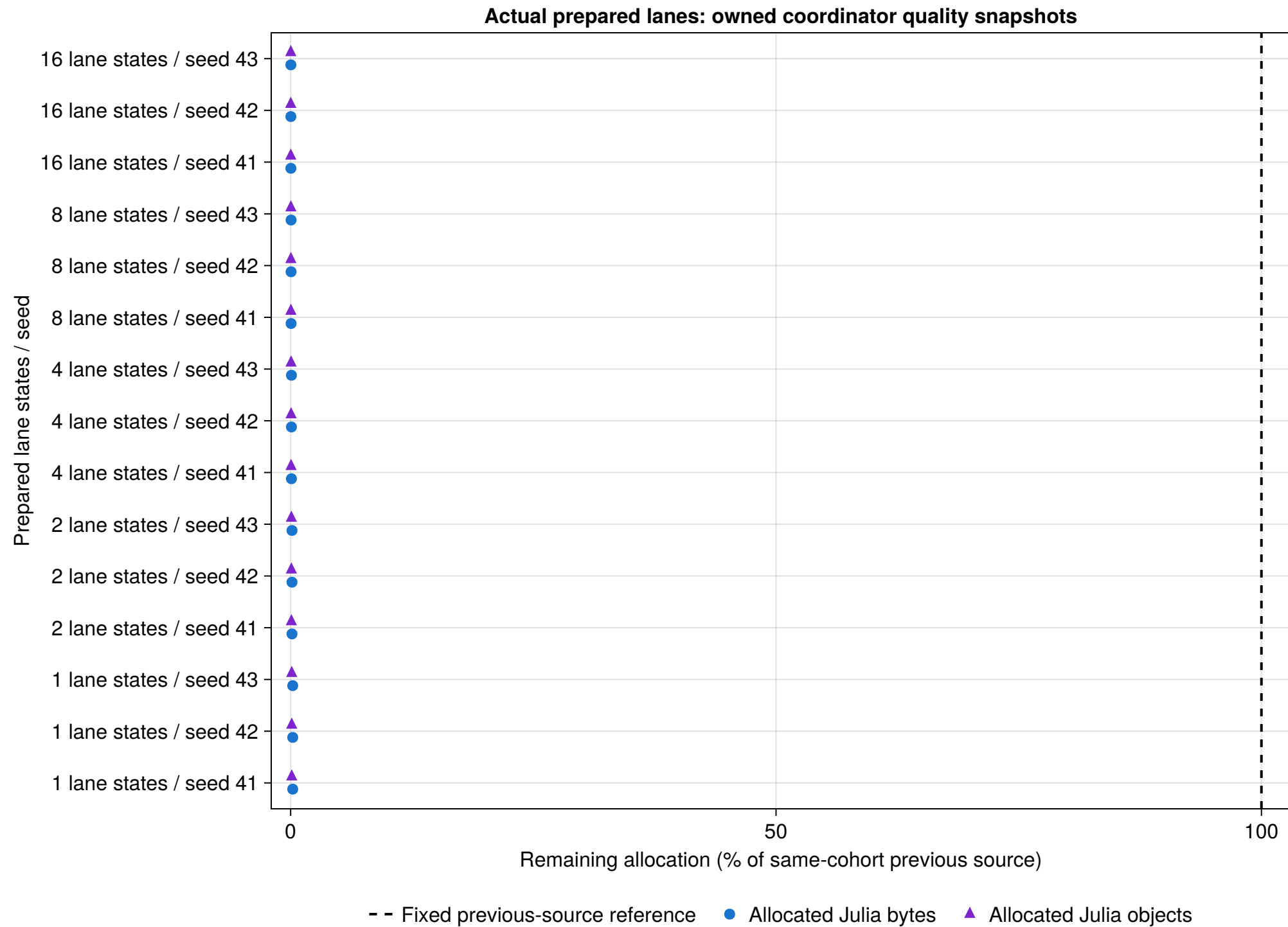




One warmed batch of 1,000 complete quality refreshes plus ordered consumption; constant 448-byte/9-object scaffold retained.
Prepared widths 1/2/4/8/16 are lane states; only 2 threads run. Qualities, checksum, lane ownership and original PDPTW solutions match.
The isolated refresh allocates zero bytes. Concurrent timing; no controlled speed or solution-quality claim.